

Question 1: Dataset Summary & Overview

Load the `employee.csv` dataset and perform basic exploratory data analysis including checking dimensions, data types, missing values, and statistical summary.

```
In [3]: import pandas as pd

# Load dataset
df = pd.read_csv("employee.csv")

# Number of rows and columns
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

# Data types
print("\nData Types:")
print(df.dtypes)

# Missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Statistical summary
print("\nStatistical Summary:")
print(df.describe(include='all'))
```

Number of Rows: 20
Number of Columns: 5

Data Types:

```
EmpID      int64
Name       object
Age        int64
Department object
Salary     float64
dtype: object
```

Missing Values:

```
EmpID      0
Name       0
Age        0
Department 0
Salary     1
dtype: int64
```

Statistical Summary:

	EmpID	Name	Age	Department	Salary
count	20.000000	20	20.000000	20	19.000000
unique	NaN	20	NaN	4	NaN
top	NaN	Emp1	NaN	HR	NaN
freq	NaN	1	NaN	5	NaN
mean	110.500000	NaN	32.600000	NaN	53578.947368
std	5.91608	NaN	8.356529	NaN	25000.701745
min	101.000000	NaN	22.000000	NaN	32000.000000
25%	105.750000	NaN	25.000000	NaN	39000.000000
50%	110.500000	NaN	30.000000	NaN	47000.000000
75%	115.250000	NaN	41.250000	NaN	56000.000000
max	120.000000	NaN	45.000000	NaN	120000.000000

Question 2: Categorical Frequency Distribution & Bar Chart

Load the `students.csv` dataset, calculate the frequency distribution of grades, and visualize it using a bar chart.

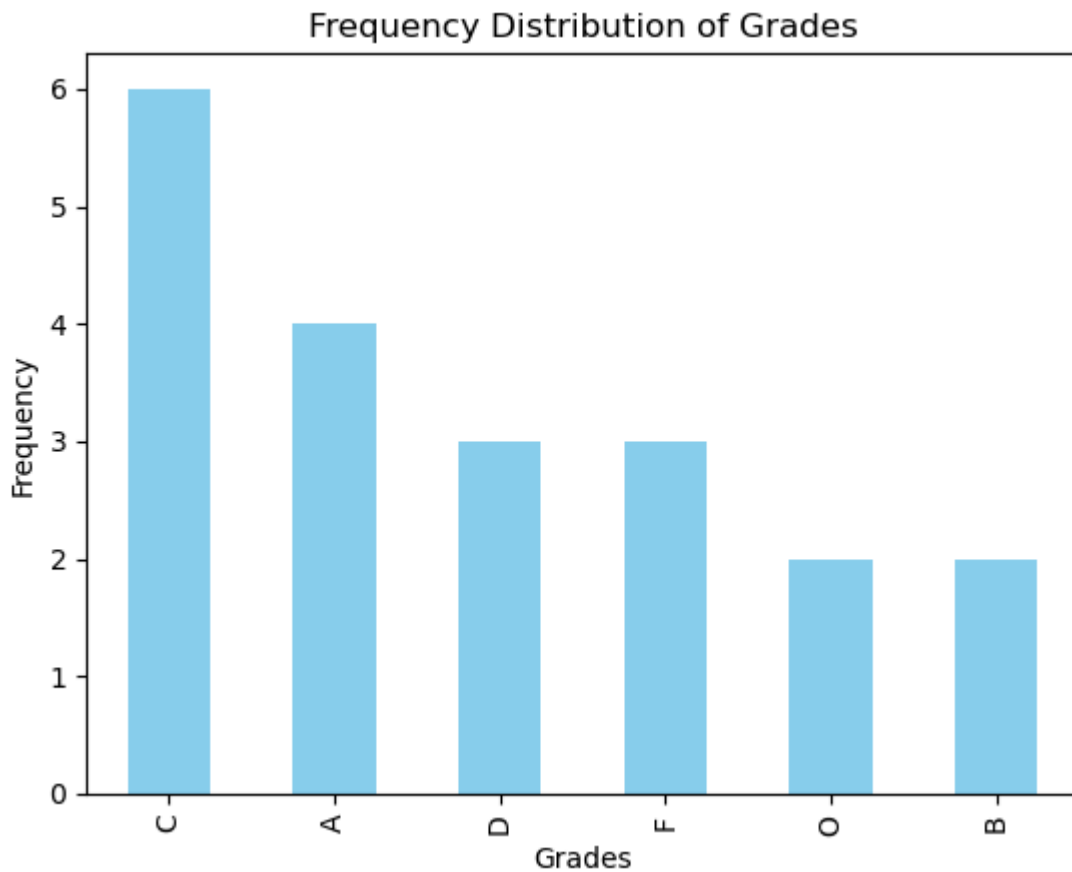
```
In [4]: import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("students.csv")

# Frequency distribution
grade_freq = df["Grade"].value_counts()
print(grade_freq)

# Bar chart
grade_freq.plot(kind="bar", color="skyblue")
plt.title("Frequency Distribution of Grades")
plt.xlabel("Grades")
plt.ylabel("Frequency")
plt.show()
```

```
Grade
C    6
A    4
D    3
F    3
O    2
B    2
Name: count, dtype: int64
```



Question 3: Histogram & Skewness Analysis

Load the `house.csv` dataset, plot a histogram for house prices, calculate skewness, and determine if the distribution is normal or skewed.

```
In [5]: import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("house.csv")

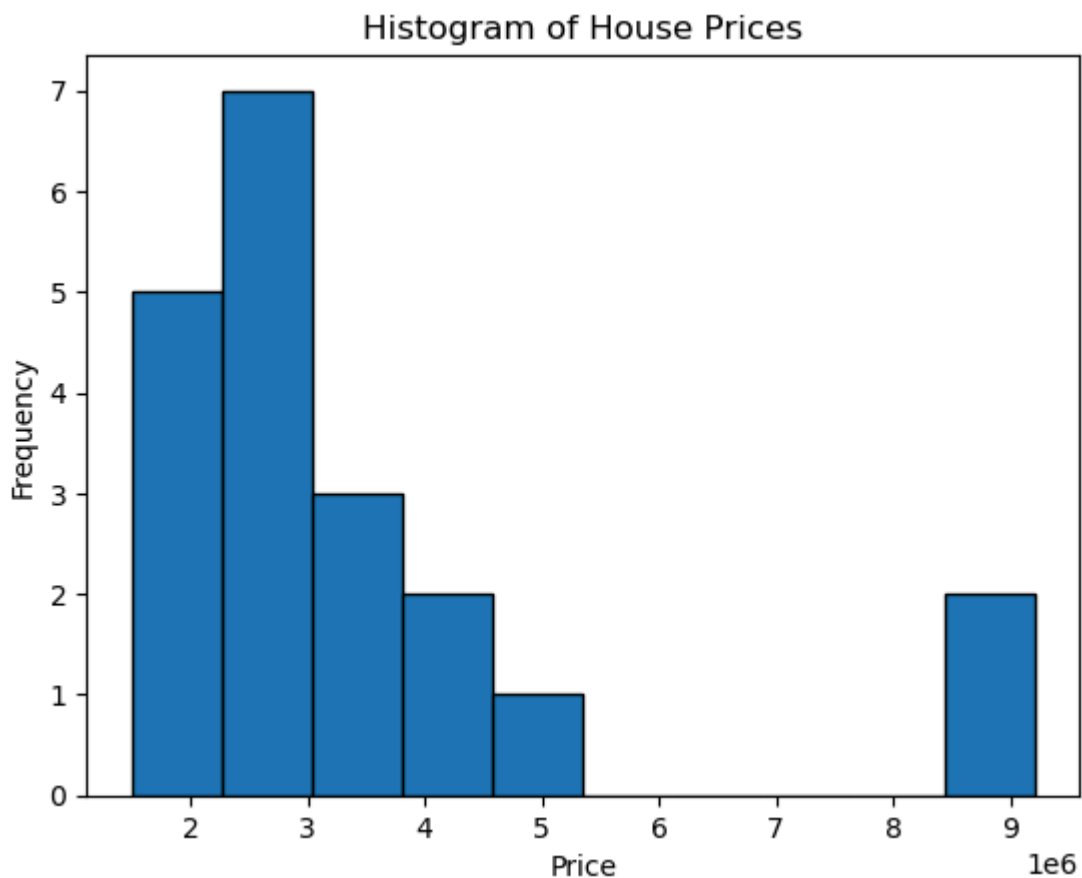
# Histogram
plt.hist(df["Price"], bins=10, edgecolor="black")
plt.title("Histogram of House Prices")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()

# Skewness
print("Skewness:", df["Price"].skew())
```

```

if abs(df["Price"].skew()) < 0.5:
    print("Approximately Normally Distributed")
else:
    print("Skewed Distribution")

```



Skewness: 2.0012770493050738
Skewed Distribution

Question 4: Box Plot & Outlier Detection (Employee Salary)

Load the `employee.csv` dataset, plot a box plot for salary, and detect outliers using the Interquartile Range (IQR) method.

```

In [6]: import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("employee.csv")

# Box plot
plt.boxplot(df["Salary"].dropna())
plt.title("Salary Box Plot")
plt.ylabel("Salary")
plt.show()

# Outliers
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

```

```

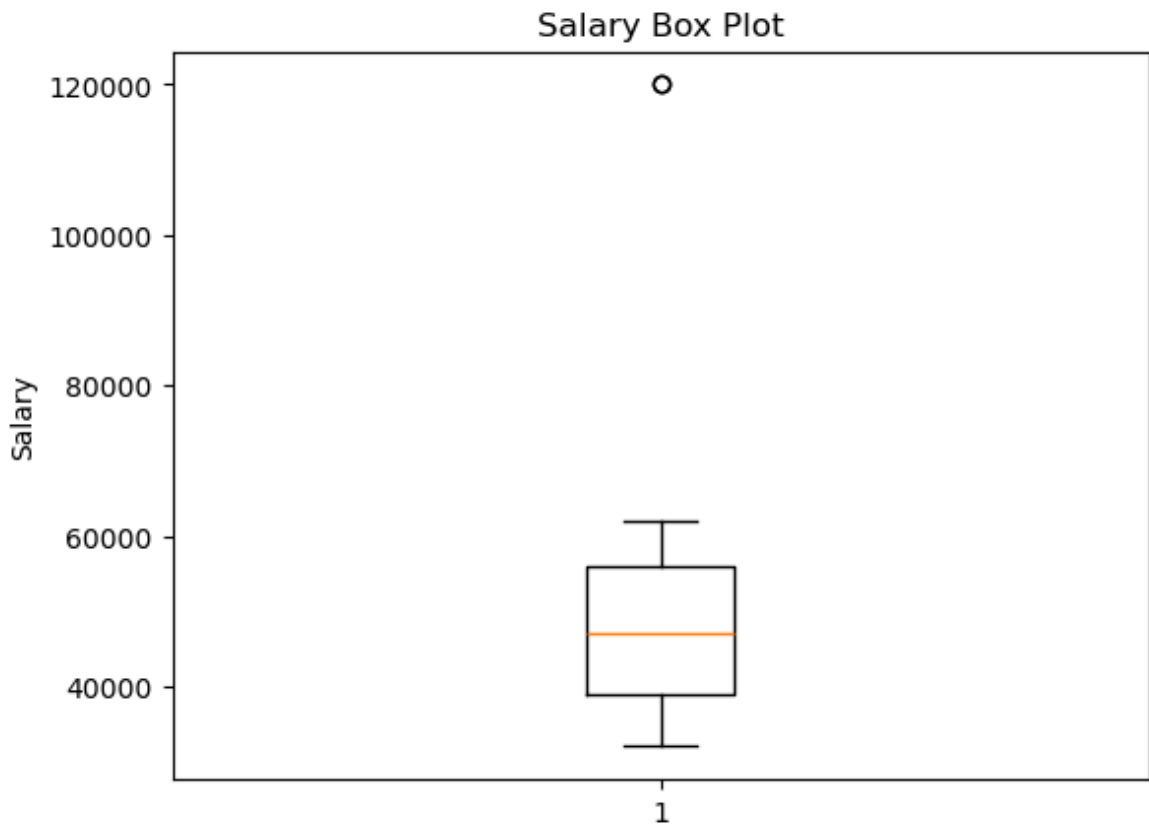
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Salary"] < lower) | (df["Salary"] > upper)]

print("Outliers:")
print(outliers)

```



Outliers:

	EmpID	Name	Age	Department	Salary
4	105	Emp5	30	IT	120000.0
8	109	Emp9	45	IT	120000.0

Question 5: Outlier Identification in Student Marks

Load the `students.csv` dataset and identify any outlier records based on student marks using the IQR method.

```

In [7]: import pandas as pd

# Load dataset
df = pd.read_csv("students.csv")

Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR

```

```
upper = Q3 + 1.5 * IQR

outliers = df[(df["Marks"] < lower) | (df["Marks"] > upper)]

print("Outliers:")
print(outliers)
```

Outliers:

Empty DataFrame

Columns: [StudentID, Name, Marks, Grade]

Index: []

Question 6: Removing Outliers from Sales Data

Load the `sales.csv` dataset and remove outliers from the `Sales` column using the IQR technique to get a cleaned dataset.

```
In [8]: import pandas as pd

# Load dataset
df = pd.read_csv("sales.csv")

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned_df = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

print("Cleaned Dataset:")
print(cleaned_df)
```

Cleaned Dataset:

	Product	Sales
0	P1	210
1	P2	220
2	P3	215
3	P4	230
4	P5	240
5	P6	250
6	P7	245
7	P8	235
8	P9	225
9	P10	260
10	P11	255
11	P12	248
12	P13	238
13	P14	242
16	P17	230
17	P18	228
18	P19	235
19	P20	240

	Product	Sales
0	P1	210
1	P2	220
2	P3	215
3	P4	230
4	P5	240
5	P6	250
6	P7	245
7	P8	235
8	P9	225
9	P10	260
10	P11	255
11	P12	248
12	P13	238
13	P14	242
16	P17	230
17	P18	228
18	P19	235
19	P20	240

Question 7: Visualization Comparison Before & After Outlier Removal

Load `sales.csv`, filter out the outliers using IQR, and plot histograms and box plots before and after outlier removal for visual comparison.

```
In [9]: import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("sales.csv")

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

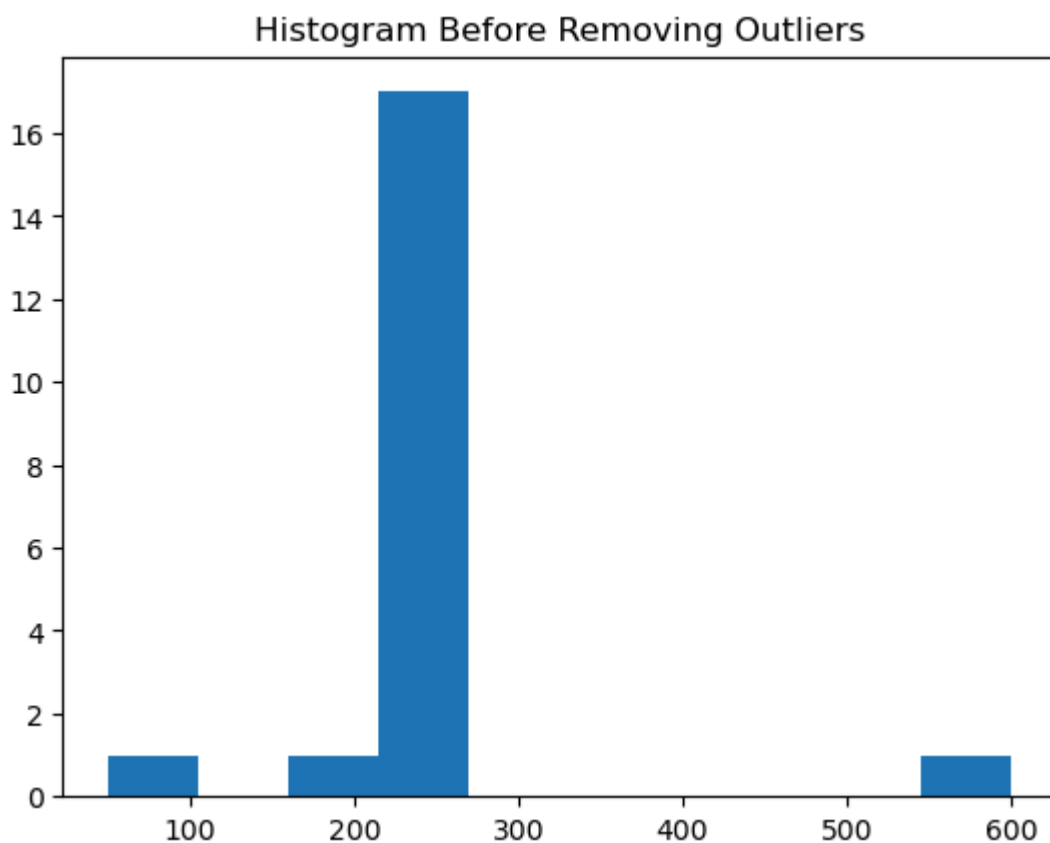
cleaned = df[(df["Sales"] >= lower) & (df["Sales"] <= upper)]

# Histogram Before
plt.hist(df["Sales"], bins=10)
plt.title("Histogram Before Removing Outliers")
plt.show()

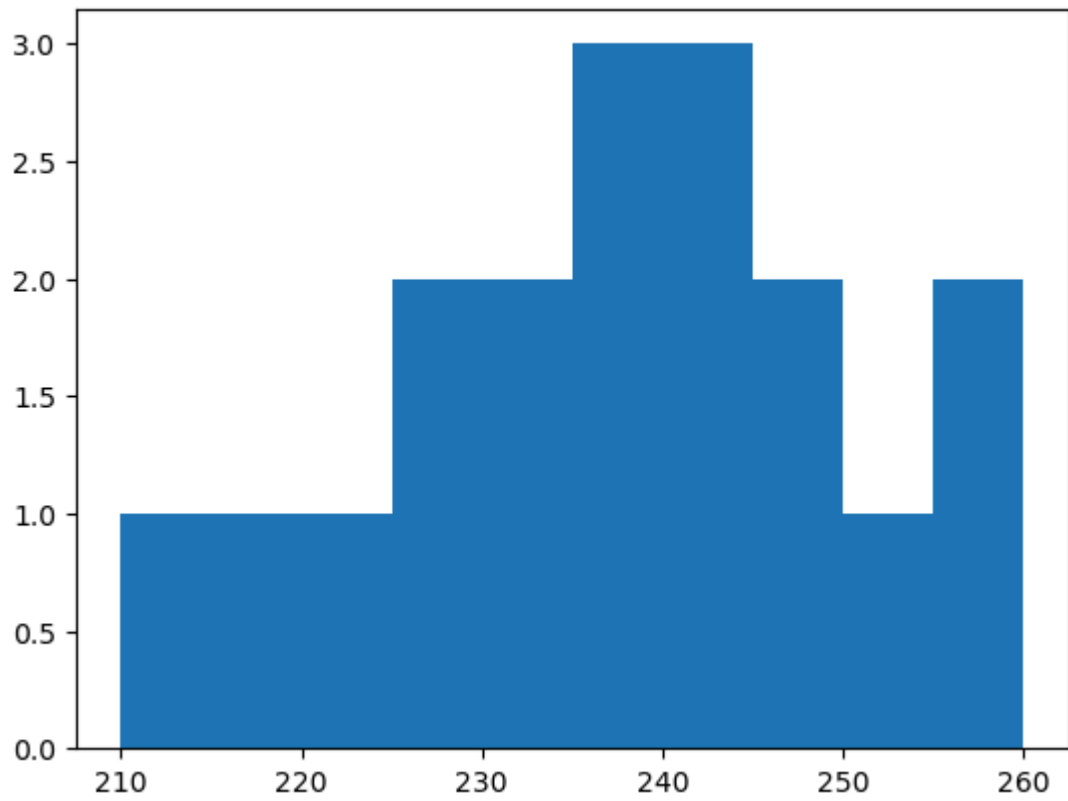
# Histogram After
plt.hist(cleaned["Sales"], bins=10)
plt.title("Histogram After Removing Outliers")
plt.show()

# Box Plot Before
plt.boxplot(df["Sales"])
plt.title("Box Plot Before Removing Outliers")
plt.show()

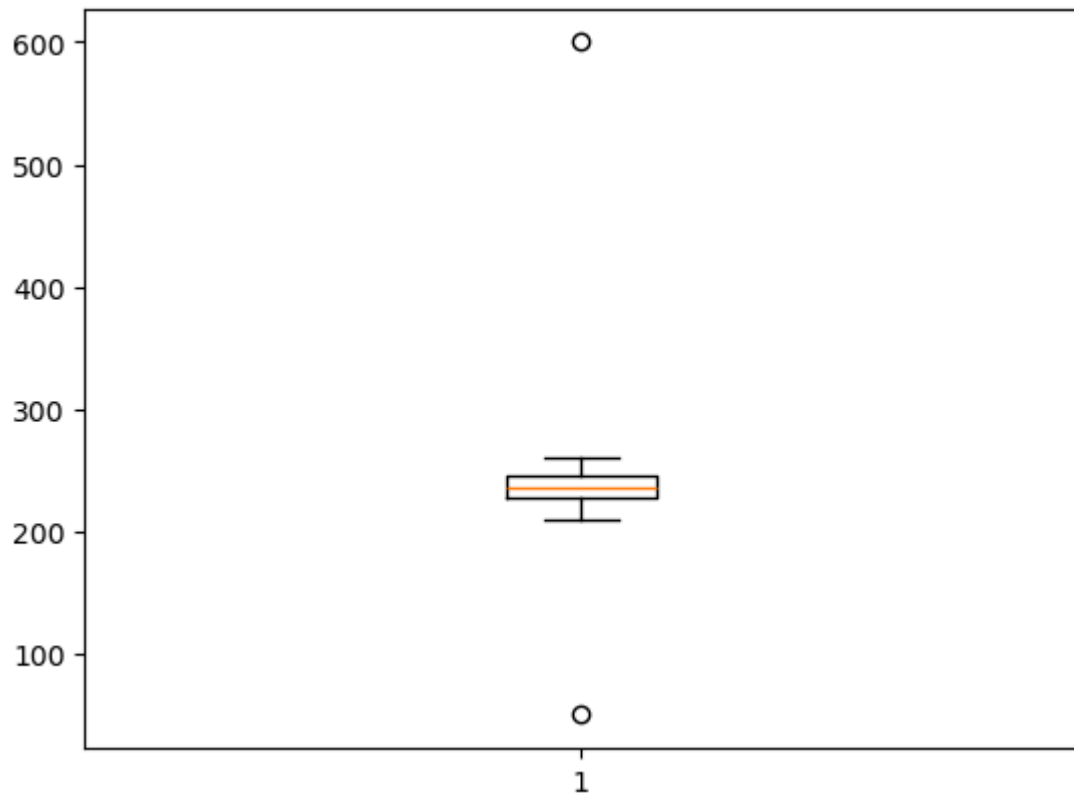
# Box Plot After
plt.boxplot(cleaned["Sales"])
plt.title("Box Plot After Removing Outliers")
plt.show()
```

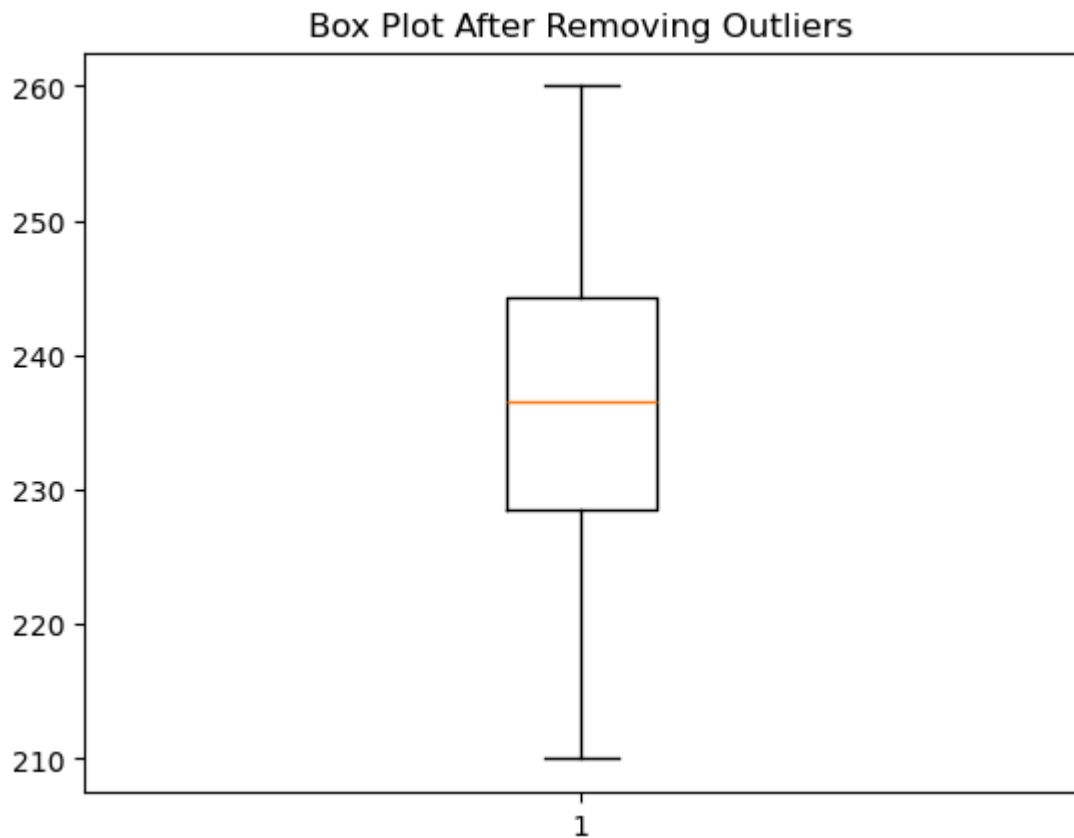


Histogram After Removing Outliers



Box Plot Before Removing Outliers





Question 8: Full Exploratory Data Analysis & Outlier Cleaning on Iris Dataset

Load the Iris dataset from `sklearn`, check missing values and descriptive statistics, plot histograms and box plots, remove numerical outliers across all features, and save the cleaned dataset.

```
In [10]: import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Load Iris dataset
iris = load_iris(as_frame=True)
df = iris.frame

# Dataset Information
print("Dataset Info:")
print(df.info())

# Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# Descriptive Statistics
print("\nDescriptive Statistics:")
print(df.describe())

# Histograms
df.hist(figsize=(8,6))
plt.show()
```

```
# Box Plots
df.plot(kind="box", figsize=(8,6))
plt.show()

# Detect and Remove Outliers
numeric_cols = df.select_dtypes(include='number').columns

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df = df[(df[col] >= lower) & (df[col] <= upper)]

print("\nCleaned Dataset:")
print(df)

# Save Cleaned Dataset
df.to_csv("iris_cleaned.csv", index=False)
print("Cleaned dataset saved as iris_cleaned.csv")
```

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal length (cm)	150 non-null	float64
1	sepal width (cm)	150 non-null	float64
2	petal length (cm)	150 non-null	float64
3	petal width (cm)	150 non-null	float64
4	target	150 non-null	int64

dtypes: float64(4), int64(1)

memory usage: 6.0 KB

None

Missing Values:

sepal length (cm) 0

sepal width (cm) 0

petal length (cm) 0

petal width (cm) 0

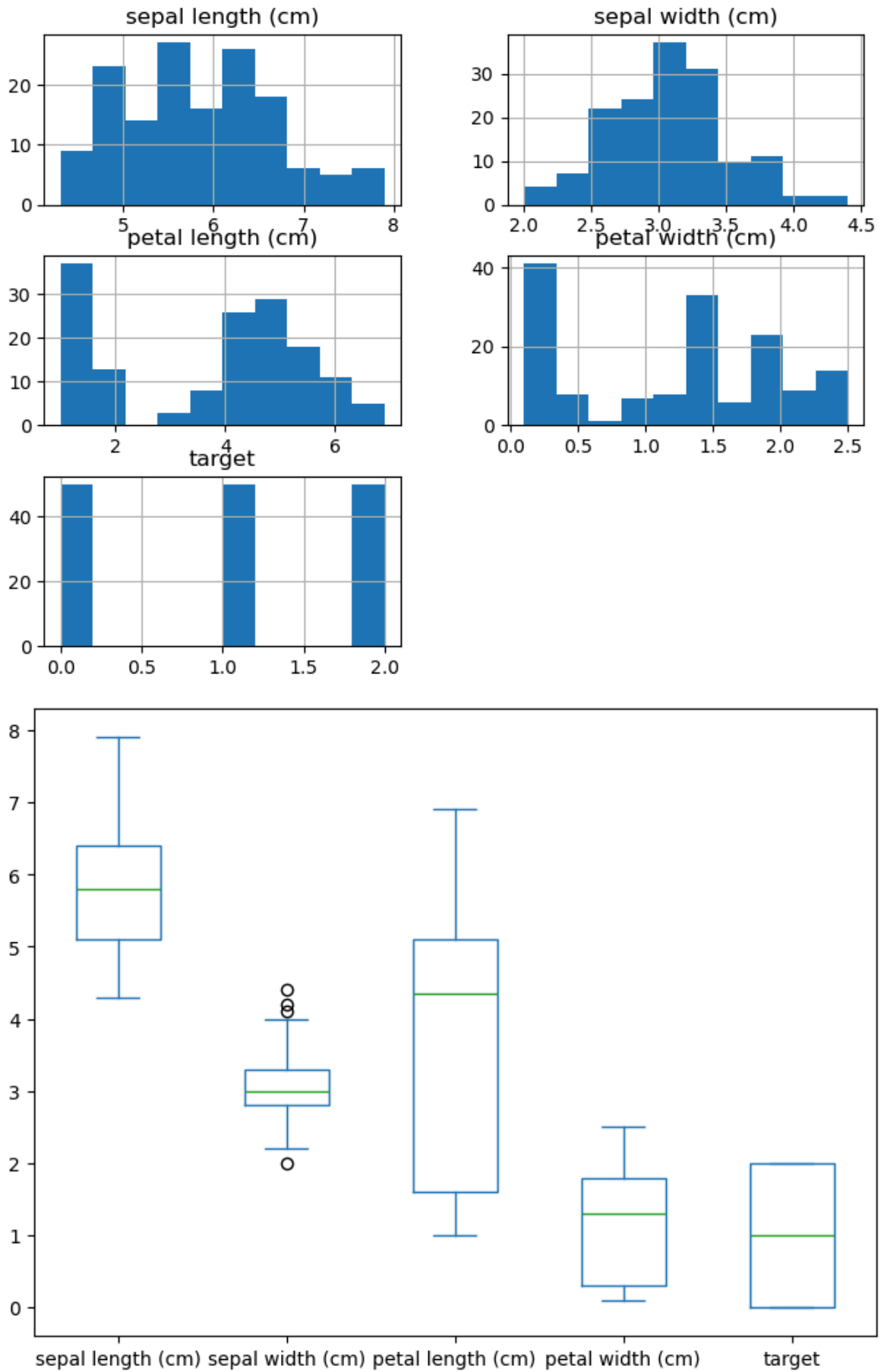
target 0

dtype: int64

Descriptive Statistics:

	sepal length (cm)	sepal width (cm)	petal length (cm) \
count	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000
std	0.828066	0.435866	1.765298
min	4.300000	2.000000	1.000000
25%	5.100000	2.800000	1.600000
50%	5.800000	3.000000	4.350000
75%	6.400000	3.300000	5.100000
max	7.900000	4.400000	6.900000

	petal width (cm)	target
count	150.000000	150.000000
mean	1.199333	1.000000
std	0.762238	0.819232
min	0.100000	0.000000
25%	0.300000	0.000000
50%	1.300000	1.000000
75%	1.800000	2.000000
max	2.500000	2.000000



Cleaned Dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	
..	...	...	...	...	
145	6.7	3.0	5.2	2.3	
146	6.3	2.5	5.0	1.9	
147	6.5	3.0	5.2	2.0	
148	6.2	3.4	5.4	2.3	
149	5.9	3.0	5.1	1.8	

	target
0	0
1	0
2	0
3	0
4	0
..	...
145	2
146	2
147	2
148	2
149	2

[146 rows x 5 columns]

Cleaned dataset saved as iris_cleaned.csv